

The packing P can also be viewed as a collection of interconnected flowers: a *flower* consists of a *central* circle C_v and the chain of successively tangent *petal* circles, $\{C_{v_0}, \dots, C_{v_{n-1}}\}$, all tangent to C_v . A flower is *closed* if $C_{v_{n-1}}$ is tangent to C_{v_0} , in which case v is an *interior* vertex of K , whereas a flower is *open* if and only if v is a *boundary* vertex. (To avoid pathologies, we require of K that every boundary vertex has at least one interior neighbor.) There are three classes of closed flowers: A *univalent* flower is one whose petals have mutually disjoint interiors. An *un-branched* flower is one whose petals wrap once around the center, possibly with overlaps between non-contiguous petals. Finally, a *branched* flower is one whose petals wrap more than once about the center and its *degree* d is number of times it wraps.

A circle packing P is *univalent* if its circles have mutually disjoint interior. It is necessary (but not sufficient that the flowers for interior vertices are univalent). If an interior circle C_v has a branched flower, then we say that P has a branch point at v .

The surprising richness of the topic is seen in the foundational existence and uniqueness result; namely, the Koebe-Andreiev-Thurston (KAT) Theorem, which states that for any triangulation K of a topological sphere, there exists an associated univalent circle packing $\mathcal{P}_K$ of the Riemann sphere $\mathbb{P}$, and that $\mathcal{P}_K$ is unique up to Möbius transformations (and inversions) of $\mathbb{P}$. Thurston also proposed in [24] a clever algorithm for actually computing such packings, allowing us today to treat circle packing as a verb: to "circle pack" a complex K is to create and manipulate associated circle packings.

1.2. On Discrete Analytic Functions. Intriguing as circle packings were in their own right, it was a conjecture in Thurston's talk that really fired up the topic. An example in Figure 1.2 will set the stage. Let P be a univalent circle packing filling a simply connected region Ω of the plane, as on the right in the figure, and let K be the underlying complex. Thurston proved using KAT that there exists a univalent circle packing for K in the unit disc $\mathbb{D}$ whose boundary circles are all horocycles, as on the left in the figure. With two circle packings for the same complex, one may define $F : \mathcal{P}_K \rightarrow P$ by identifying corresponding circles. Essentially a mapping from the unit disc to Ω , F is roughly analogous to the classical Riemann Mapping.

It is the conjecture Thurston made about such discrete conformal mappings that kicked off the nearly 40 years of development efforts in circle packing. He suggested that if one were to refine this construction — used circle packings P with ever more and smaller circles — that the